AN INTERSECTION OF QUASI-COMPACT OPENS IN AN IND-QUASI-AFFINE SCHEME IDENTIFIES WITH AN INTERSECTION OF QUASI-COMPACT OPENS IN AN AFFINE SCHEME, AND OPENS OF X INSIDE E MAP TO OPENS OF Y

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Allowed standard prerequisites (editorial).

- The canonical morphism from a locally ringed space with a section-generating global sections map into the spectrum of its ring of sections (schemes chapter, Lemma schemes-lemma-morphism-into-affine, canonical tag 01I1). cited at statement lines 3215-3219 and proof lines 3227-3229. Produces the canonical morphism j : (E, O_X|_E) to (Spec(A), O_Spec(A)) whose image identification is the content of the lemma.
- Anti-equivalence of colimits of rings and limits of affine schemes on underlying topological spaces (limits chapter, Lemma limits-lemma-topology-limit, canonical tag 01YY). cited at proof lines 3244-3245. From A = colim A_i it gives Spec(A) = lim Spec(A_i) as topological spaces, the limit through which the image E' is computed.
- Presentation of a spectral-like intersection as a limit of schemes (topology chapter, Lemma topology-lemma-make-spectral-space, canonical tag 0A31). cited at proof lines 3246-3247. Gives E = lim U_i, which combined with the limit identification shows E maps homeomorphically onto its image E' inside Spec(A).
- Cartesianity of the square attached to a quasi-compact open of a quasi-affine scheme (properties chapter, Lemma properties-lemma-cartesian-diagram-quasi-affine, canonical tag 0ARY). cited at proof lines 3273-3275. Makes the first of the three squares cartesian, hence the second, and after passing to the limit the third, whose top horizontal arrow is then an open immersion.

Source reference. Original lemma 01I1: sources/stacks-chow/tags.txt, source lines 1962–1962. Cited at statement lines 3215-3219 for the canonical morphism j and at proof lines 3227-3229 for its applicability to the locally ringed space (E, O_X|_E) with A = Gamma(E, O_X|_E).

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Source reference. Original lemma 01YY: sources/stacks-chow/tags.txt, source lines 2573–2573. Cited at proof lines 3244-3245: A = colim A_i gives Spec(A) = lim Spec(A_i) as topological spaces.

Source reference. Original lemma 0A31: sources/stacks-chow/tags.txt, source lines 12724–12724. Cited at proof lines 3246-3247: E = lim U_i, the input for the homeomorphism of E onto its image in Spec(A).

Source reference. Original lemma 0ARY: sources/stacks-chow/tags.txt, source lines 13561–13561. Cited at proof lines 3273-3275: the first square is cartesian, hence the second, and the limit square whose top arrow is an open immersion.

Lemma 0.2. *Let X be an ind-quasi-affine scheme. Let $E \subset X$ be an intersection of a nonempty family of quasi-compact opens of X. Set $A = \Gamma(E, \mathcal{O}_X|_E)$ and $Y = \mathrm{Spec}(A)$. Then the canonical morphism*

$$j : (E, \mathcal{O}_X|_E) \longrightarrow (Y, \mathcal{O}_Y)$$

of Schemes, Lemma 01I1 determines an isomorphism $(E, \mathcal{O}_X|_E) \to (E', \mathcal{O}_Y|_{E'})$ where $E' \subset Y$ is an intersection of quasi-compact opens. If $W \subset E$ is open in X, then $j(W)$ is open in Y.

Proof. Note that $(E, \mathcal{O}_X|_E)$ is a locally ringed space so that Schemes, Lemma 01I1 applies to $A \to \Gamma(E, \mathcal{O}_X|_E)$. Write $E = \bigcap_{i \in I} U_i$ with $I \neq \emptyset$ and $U_i \subset X$ quasi-compact open. We may and do assume that for $i, j \in I$ there exists a $k \in I$ with $U_k \subset U_i \cap U_j$. Set $A_i = \Gamma(U_i, \mathcal{O}_{U_i})$. We obtain commutative diagrams

$$\begin{array}{ccc} (E, \mathcal{O}_X|_E) & \longrightarrow & (\mathrm{Spec}(A), \mathcal{O}_{\mathrm{Spec}(A)}) \\ \downarrow & & \downarrow \\ (U_i, \mathcal{O}_{U_i}) & \longrightarrow & (\mathrm{Spec}(A_i), \mathcal{O}_{\mathrm{Spec}(A_i)}) \end{array}$$

Since U_i is quasi-affine, we see that $U_i \to \mathrm{Spec}(A_i)$ is a quasi-compact open immersion. On the other hand $A = \mathrm{colim}\, A_i$. Hence $\mathrm{Spec}(A) = \lim \mathrm{Spec}(A_i)$ as topological spaces (Limits, Lemma 01YY). Since $E = \lim U_i$ (by Topology, Lemma 0A31) we see that $E \to \mathrm{Spec}(A)$ is a homeomorphism onto its image E' and that E' is the intersection of the inverse images of the opens $U_i \subset \mathrm{Spec}(A_i)$ in $\mathrm{Spec}(A)$. For any $e \in E$ the local ring $\mathcal{O}_{X,e}$ is the value of $\mathcal{O}_{U_i,e}$ which is the same as the value on $\mathrm{Spec}(A)$.

To prove the final assertion of the lemma we argue as follows. Pick $i, j \in I$ with $U_i \subset U_j$. Consider the following commutative diagrams

$$\begin{array}{ccc} U_i & \longrightarrow & \mathrm{Spec}(A_i) \\ \downarrow & & \downarrow \\ U_i & \longrightarrow & \mathrm{Spec}(A_j) \end{array} \qquad \begin{array}{ccc} W & \longrightarrow & \mathrm{Spec}(A_i) \\ \downarrow & & \downarrow \\ W & \longrightarrow & \mathrm{Spec}(A_j) \end{array} \qquad \begin{array}{ccc} W & \longrightarrow & \mathrm{Spec}(A) \\ \downarrow & & \downarrow \\ W & \longrightarrow & \mathrm{Spec}(A_j) \end{array}$$

By Properties, Lemma 0ARY the first diagram is cartesian. Hence the second is cartesian as well. Passing to the limit we find that the third diagram is cartesian, so the top horizontal arrow of this diagram is an open immersion. □